

Applying Percentages

Section One: Finding the Percentage of a Quantity

IMPORTANT

$$\text{Percentage of a quantity} = \frac{\text{Percentage}}{100} \times \text{quantity} \text{ OR } \text{Percentage} \div 100 \times \text{quantity}$$

- 4 Joanna earns \$97.50 per week from her part-time job. She decides to save 30% of her income so she can go on holiday at the end of Year 12.
- How much does she save each week? Round your answer to the nearest dollar.
 - Joanna is given a 2.5% wage increase. How much extra does she earn each week? Round your answer to the nearest cent.

$$\begin{aligned} \text{a) } & \frac{30}{100} \times \$97.50 \\ & = \$29.25 \\ & \approx \$29 \end{aligned}$$

Joanna needs to save \$29 per week.

$$\begin{aligned} \text{b) } & \frac{2.5}{100} \times \$97.50 \\ & = \$2.4375 \\ & \approx \$2.44 \end{aligned}$$

Joanna has a pay rise of \$2.44 per week.

- 5 Brandon pays \$225 per week in rent for his share of a house. The landlord decides to increase the rent by 4%.
- How much extra rent will Brandon have to pay?
 - What is the total amount of rent that he will have to pay?

$$\begin{aligned} \text{a) } & \frac{4}{100} \times \$225 \\ & = \$9 \end{aligned}$$

Brandon's rent increased by \$9 per week.

$$\text{b) } \$225 + \$9 = \$234$$

Brandon's new rent payment is \$234 per week.

- 8 Stamp duty is a tax paid to the state government when you purchase a new car. In Tasmania, Bruce will pay 3% of the purchase price in stamp duty. He buys a new car for \$29 990. How much stamp duty will he pay?

$$\frac{3}{100} \times \$29\,990 = \$899.70$$

Bruce will need to pay \$899.70 in stamp duty.

- 10 Australia has approximately 800 000 km of roads. Approximately 40.5% of Australian roads are sealed with bitumen or concrete. All other roads are gravel, sand or dirt.
- How many kilometres of Australian roads are sealed?
 - Of the roads that are NOT sealed, approximately 66% are improved surfaces. How many kilometres of improved surface roads are there?

$$\begin{aligned} \text{a) } & \frac{40.5}{100} \times 800\,000 \\ & = 324\,000 \text{ km of} \\ & \text{sealed road in} \\ & \text{Australia} \end{aligned}$$

$$\begin{aligned} \text{b) } & \frac{66}{100} \times 800\,000 \\ & = 528\,000 \text{ km of} \\ & \text{improved surfaces in} \\ & \text{Australia} \end{aligned}$$

Section Two: Comparing quantities using percentages

IMPORTANT

$$\text{Percentage} = \frac{\text{amount}}{\text{whole amount}} \times 100\% \quad \text{OR} \quad \text{amount} \div \text{whole amount} \times 100\%$$

- 3 In a match, basketball players rarely have the same number of shots at goal. Over the last three matches, Brian was successful with 18 shots out of 33 attempts. Over the same three matches, Luke was successful with 23 shots out of 37 attempts.

Which player is more successful? Justify your answer.

$$\text{Brian } \frac{18}{33} \times 100 = 54.5\% \text{ successful attempts}$$

$$\text{Luke} = \frac{23}{37} \times 100 = 62.2\% \text{ successful attempts}$$

Luke is more successful because he scored 7.7% more goals than Brian.

- 5 Colombia has an area of 1 141 748 km², of which 100 210 km² is water. Iran has an area of 1 648 195 km², of which 116 600 km² is water.
- Calculate the percentage of water for each country. Give your answer correct to 2 decimal places.
 - Which country has a greater percentage of water?
 - Why might the percentage of water be important?

$$\text{a) Colombia } \frac{100210}{1141748} \times 100 = 8.78\% \text{ water}$$

$$\text{Iran} = \frac{116600}{1648195} \times 100 = 7.07\% \text{ water}$$

b) Colombia has the highest percentage of water.

c) Agriculture, industry, people
Water is essential for life.

- 6 **Example 3** Write the first quantity as a percentage of the second. Remember to express both quantities in the same units first.

a 3 minutes, 1 hour

b 75 mm, 20 cm

c 400 g, 32 kg

d 18 hours, 3 days

$$\text{a) } 1 \text{ hr} = 60 \text{ min}$$

$$\frac{3}{60} \times 100 = 5\%$$

$$\text{c) } 32 \text{ kg} = 32000 \text{ g}$$

$$\frac{400}{32000} \times 100 = 1.25\%$$

$$\text{b) } 20 \text{ cm} = 200 \text{ mm}$$

$$\frac{75}{200} \times 100 = 37.5\%$$

$$\text{d) } 3 \text{ days} = 72 \text{ hours}$$

$$\frac{18}{72} \times 100 = 25\%$$